



S^3 AD: Efficient industrial anomaly detection via selection, space, and scale

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ABSTRACT

Real-world industrial visual anomaly detection (VAD) demands the simultaneous handling of diverse product categories under strict real-time constraints. However, the self-attention mechanism inherent in standard Vision Transformers (ViTs) incurs quadratic computational complexity during global context modeling, thereby limiting their real-time applicability in VAD. Crucially, this bottleneck is largely driven by the uniform processing of massive background redundancy common in industrial scenarios. To bypass this inefficiency while retaining high precision, we propose S^3 AD, a unified framework anchored in Selection, Space, and Scale. Instead of dense global interactions, S^3 AD employs a Saliency-Guided Sparse Attention to focus resources solely on salient areas. Specifically, regarding Selection, we devise a discriminating Top-K pruning strategy guided by a hybrid metric of attention scores and feature magnitudes, which filters out non-informative tokens to minimize cost. Simultaneously, for Space, we incorporate Geometry-Preserving Positional Embeddings to explicitly anchor the selected tokens, ensuring structural integrity is maintained despite sparse interaction. Finally, concerning Scale, we integrate an Exchangeable Feature Pyramid Network (E-FPN) to ensure rigorous feature fusion across resolutions. Extensive experiments demonstrate that S^3 AD achieves state-of-the-art performance and establishes a superior accuracy–efficiency Pareto frontier, consistently surpassing strong baselines on multiple benchmarks. The code is available at: <https://github.com/CreatedTRYNA/S3AD>.

1. Introduction

Industrial visual anomaly detection (VAD) is a cornerstone technology in intelligent manufacturing [1]. Modern production lines increasingly demand unified multi-class detection, posing significant challenges due to diverse inter-class distributions and complex defect patterns [2]. Consequently, practical VAD models face the dual challenge of robustly capturing global contextual information to distinguish anomalies while strictly adhering to real-time inference constraints [3].

To address these challenges, recent research has largely explored State Space Models (SSMs) and Vision Transformers (ViTs). While SSMs like the MambaAD family [4,5] achieve linear inference complexity, they potentially exhibit limitations in the precise long-range feature alignment critical for identifying subtle defects requiring strict point-to-point correspondences [6,7]. Conversely, ViT-based architectures demonstrate exceptional robustness. Methods like UniAD [8] and ViTAD [9] leverage explicit self-attention to achieve competitive accuracy on benchmarks like MVTec AD [10] and VisA [11]. These findings highlight that explicit attention remains a powerful tool for capturing the fine-grained details required in industrial scenarios.

However, relying solely on standard ViTs introduces a significant bottleneck: computational redundancy (Fig. 1(a)). Treating all tokens equally applies expensive global attention uniformly, which is highly inefficient since most industrial image regions are non-anomalous backgrounds. Performing quadratic calculations on these information-sparse areas increases inference latency without proportional performance gains [12,13]. Specifically, to meet strict industrial requirements (e.g., > 30 FPS), recent methods often shrink the network architecture. However, this compromise degrades the detection of subtle, high-impact defects on complex datasets like VisA, MPDD, and Real-IAD, highlighting the urgent need to mitigate background redundancy without sacrificing representational capacity.

To address this, we propose S^3 AD, a unified framework balancing expressive global modeling and efficient inference. Anchored in the principles of Selection, Space, and Scale, S^3 AD shifts from dense global computation to adaptive sparse focusing. Specifically, we employ a Selection strategy to prune background redundancy, introduce Space-based geometric anchors to compensate for structural awareness lost

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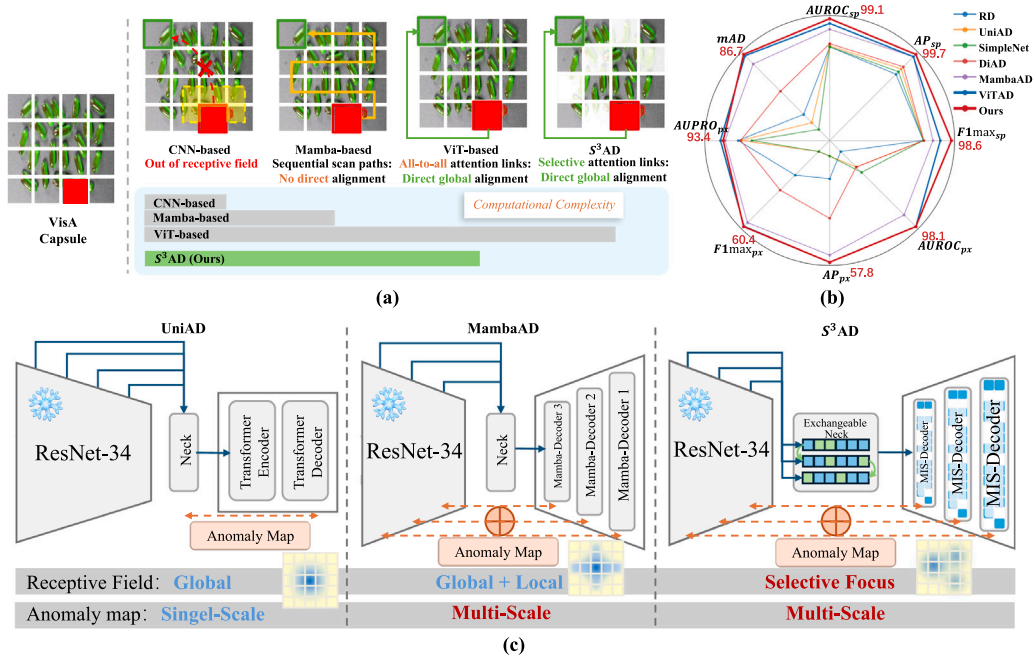


Fig. 1. The Proposed S^3AD Framework and Model Comparisons. (a) Unlike existing architectures that face a trade-off between speed and performance, S^3AD achieves an effective compromise by combining ViT's global alignment with selective feature modeling. (b) It achieves state-of-the-art performance on MVTEC AD while maintaining efficient inference speeds. (c) Architecture overview and comparison with representative works.

during sparse sampling, and utilize a parameter-efficient Scale exchange mechanism to resolve semantic confusion in traditional fusion networks.

Technically, we introduce a Saliency-Guided Sparse Attention (SGSA) mechanism to instantiate the Selection dimension. SGSA employs dynamic token selection driven by a hybrid metric of attention scores and feature magnitudes, focusing computational resources on information-dense regions while pruning redundant background tokens. To ensure robustness during this sparse interaction, we incorporate Geometry-Preserving Positional Embeddings to explicitly anchor selected tokens to their spatial contexts. Furthermore, we integrate an Exchangeable Feature Pyramid Network (E-FPN) [14] in the neck, utilizing a parameter-efficient channel exchange strategy to maximize multi-scale feature utilization without additional overhead.

The main contributions of this work are as follows:

- We propose S^3AD , a unified VAD framework anchored in Selection, Space, and Scale, successfully reconciling the conflict between expressive global context modeling and strict industrial efficiency requirements.
- We devise the Saliency-Guided Sparse Attention (SGSA) mechanism to dynamically filter non-informative background tokens via a hybrid metric, minimizing computational overhead without compromising precision on salient regions.
- We integrate Geometry-Preserving Positional Embeddings and an Exchangeable FPN to explicitly anchor selected tokens to their spatial contexts and facilitate parameter-efficient multi-scale feature fusion despite sparse interactions.
- Extensive experiments on four benchmarks show S^3AD outperforms ViT and Mamba baselines, establishing a superior accuracy-efficiency Pareto frontier.

2. Related work

2.1. Visual anomaly detection

Visual anomaly detection (VAD) is pivotal for industrial and medical inspection [15,16]. Existing approaches generally fall into three

paradigms. Augmentation-based methods, such as RealNet [17], SaliencyCut [18] and SimpleNet [19], synthesize artificial anomalies for discriminative training. Embedding-based approaches model normal distributions in pre-trained feature spaces using normalizing flows [20] or memory banks [13,21]. Finally, reconstruction-based methods like ViTAD [9] and UniVAD [22] assume models trained on normal data fail to reconstruct anomalies, utilizing residual errors and attention mechanisms to localize defects.

To mitigate the overhead of the traditional one-model-per-class paradigm, Multi-class Unsupervised Anomaly Detection (MUAD) learns a single unified model for diverse categories, improving parameter efficiency and scalability. Recent advances primarily involve generative frameworks like UniAD [8], DIAD [23], DecAD [24], and OmniAD [25] utilizing unified reconstruction or diffusion, alongside Transformer-based architectures like ViTAD [9] and DFGR [26] that exploit global modeling for cross-domain detection. Evaluations on benchmarks like ADer [27] and Real-IAD [28] confirm MUAD as a promising, scalable pathway for real-world inspection.

2.2. Long-range dependency modeling in VAD

ViTs are dominant in VAD due to their global relationship modeling. Methods like UniAD and ViTAD [8,9] leverage explicit self-attention for state-of-the-art localization. However, the quadratic complexity of self-attention limits real-time scalability. Consequently, State Space Models like MambaAD and Wave-MambaAD [4,5] have emerged as linear-time alternatives, utilizing selective updates to reduce overhead while preserving long-range modeling.

Recent advancements also leverage pre-trained Foundation Models for zero-shot and few-shot VAD. Pure vision models like DINOv2 [29] and v3 [30] extract robust patch-level representations. Works such as Dinomaly [31], INP-Former [32], and AD-DINOv3 [33] utilize these features to model intrinsic normal patterns and enhance spatial context without external references. Concurrently, Vision-Language Models like CLIP capture cross-modal contexts. Methods including Anomaly-CLIP [34], Ada-CLIP [35], Bayes-PFL [36], and AA-CLIP [37] align visual features with text prompts to distinguish abnormalities. However, these foundation models inherently suffer from exceptionally large sizes and intensive computational costs.

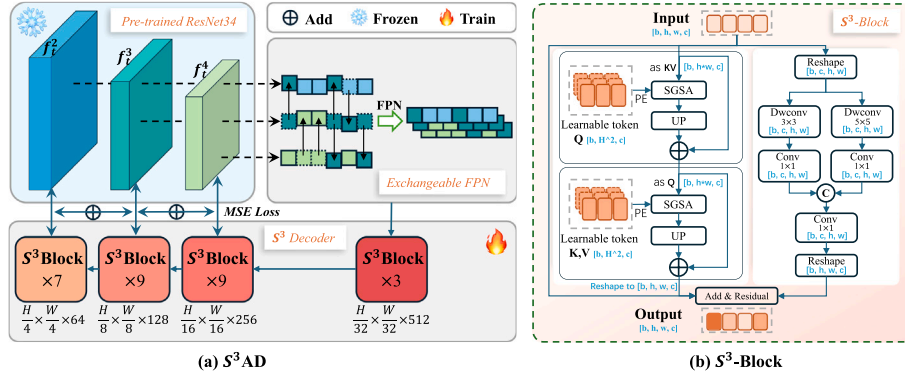


Fig. 2. Detailed architecture of the proposed S^3AD framework. (a) Overview of the overall pipeline. (b) Detailed structure of the S^3 -Block. Let b, h, w, c , and H denote the batch size, spatial height, spatial width, channel dimension, and number of attention heads, respectively. Within the attention mechanism, Q, K, V for learnable tokens are zero-initialized parameters, while those for image tokens are linearly projected from spatial features.

2.3. Token pruning and sparse attention

To alleviate the quadratic computational overhead of ViTs, Top-K token reduction techniques are widely adopted via pruning or merging. DynamicViT [38] and EViT [39] prune uninformative tokens based on global attention. Alternatively, ToMe [40] merges visually redundant features using bipartite matching. Recent architectures like QuickMerge++ [41] and ViTCoP [42] introduce feature norms or frequency priors as advanced proxies. However, these methods primarily prioritize components aligned with global macroscopic semantics, excelling mainly in macroscopic classification tasks.

However, adapting these Top-K mechanisms directly to MUAD often presents significant challenges. Because industrial anomalies are frequently minute and highly localized, tokens containing crucial defect information may yield relatively low global attention scores compared to dominant normal patterns. Consequently, they risk being inadvertently discarded as background noise by attention-based methods like EViT, or improperly merged by similarity-based methods such as ToMe, potentially compromising critical reconstruction details early on. To address this limitation, we propose SGSA. By jointly evaluating global attention saliency and local information density via a hybrid scoring paradigm, SGSA ensures that subtle anomalies are robustly preserved. This approach effectively filters redundant backgrounds while maintaining the essential informational fidelity required for MUAD.

3. Method

3.1. Overview of S^3AD

To enhance global context modeling while minimizing computational overhead, we propose the S^3AD framework anchored in Selection, Space, and Scale (Fig. 2).

During training, S^3AD operates under a teacher-student distillation paradigm. Given an input image X , a frozen Teacher network like ResNet-34 [43] extracts multi-scale features from stages 2 through 4. The Exchangeable Feature Pyramid Network (E-FPN, Section 3.4) aligns and fuses these into a unified semantic prior, ensuring rigorous cross-resolution fusion (Scale). The Student network then processes these synthesized features through stacked S^3 -Blocks. Within each block, the Saliency-Guided Sparse Attention (SGSA, Section 3.2) mechanism dynamically filters non-informative tokens, focusing computational resources on salient regions (Selection). Concurrently, Geometry-Preserving Positional Embeddings (Section 3.3) explicitly anchor selected tokens to their spatial contexts, maintaining structural integrity despite sparse interactions (Space). A Smoothing Strategy

Algorithm 1 Overall pipeline of S^3AD .

Input: Training set $\mathcal{D}_{train}$, Test set $\mathcal{D}_{test}$; Teacher $T(\cdot)$, Student $S(\cdot)$, E-FPN, SGSA, GPE, Decoder $D(\cdot)$.

Output: Anomaly heatmap $\mathcal{M}$, Binary mask $\mathcal{B}$.

1: **Training Phase:**

2: **while** not converged **do**

3: Sample batch $x \sim \mathcal{D}_{train}$

4: $\{f_t^l\} \leftarrow T(x)$

▷ Extract Teacher Features

5: $s \leftarrow E\text{-FPN}(\{f_t^l\})$

▷ Scale: Feature Fusion

6: $z \leftarrow \text{SGSA}(s) + \text{GPE}$

▷ Selection & Space

7: $\{f_s^l\} \leftarrow D(z)$

▷ Reconstruction

8: Update parameters by minimizing $\sum_l \|f_s^l - f_t^l\|_2^2$

9: **end while**

10: **Inference Phase:**

11: **for** each test image $x \in \mathcal{D}_{test}$ **do**

12: Obtain $\{f_s^l\}, \{f_t^l\}$ via forward pass

13: $\mathcal{M} \leftarrow \sum \text{Upsample}(\|f_s^l - f_t^l\|_2)$

14: $\mathcal{B} \leftarrow (\mathcal{M} > \tau)$

15: **end for**

16: **Return** $\mathcal{M}, \mathcal{B}$

(Section 3.2.1) is applied to mitigate instability from aggressive token removal. Finally, a decoder progressively upsamples the features, optimized by the Mean Squared Error (MSE) loss function:

$$\mathcal{L}_{\text{MSE}} = \sum_l \|\mathbf{f}_s^l - \mathbf{f}_t^l\|_2^2, \quad (1)$$

where $\mathbf{f}_t^l$ and $\mathbf{f}_s^l$ denote the features from the teacher and student networks at scale l , respectively.

During inference, the model quantifies anomalies using the feature discrepancy between the teacher and student networks. Pixel-wise error maps are computed, bilinearly upsampled, and aggregated to generate the final anomaly heatmap, directly indicating defect locations and shapes. The overall end-to-end workflow is summarized in Algorithm 1. Core model components are detailed subsequently.

3.2. Saliency-guided sparse attention

We detail the Saliency-Guided Sparse Attention (SGSA) mechanism. SGSA preserves explicit attention at salient regions for discriminative capacity and employs selective computation elsewhere to curtail complexity, effectively balancing expressive power and efficiency.

As shown in Fig. 3a, given input query, key, and value tokens $\mathbf{Q}_{in}, \mathbf{K}_{in}, \mathbf{V}_{in} \in \mathbb{R}^{B \times N \times C}$, we apply linear projections to obtain $\mathbf{Q}, \mathbf{K}, \mathbf{V}$:

$$\mathbf{Q} = \mathbf{Q}_{in} \mathbf{W}^Q, \quad \mathbf{K} = \mathbf{K}_{in} \mathbf{W}^K, \quad \mathbf{V} = \mathbf{V}_{in} \mathbf{W}^V, \quad (2)$$

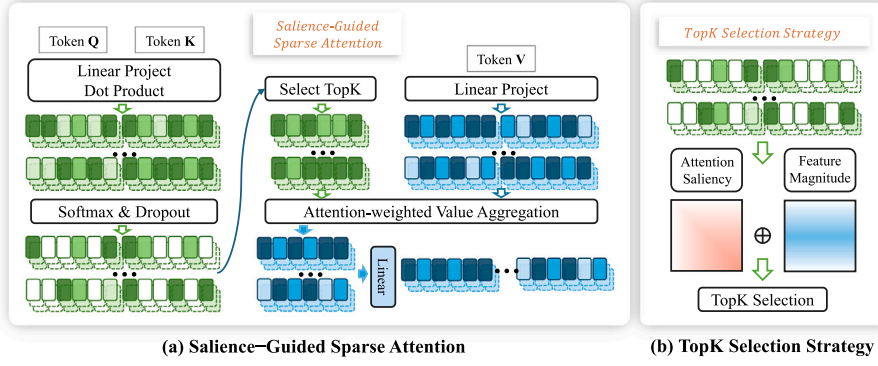


Fig. 3. Overall structure of the SGSA.(a): SGSA structure preserving explicit attention at key locations while performing selective updates elsewhere to reduce cost. (b): Top-K selection strategy ranking tokens via a hybrid metric of attention scores and feature magnitudes to retain only informative tokens.

where $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{C \times C}$ are projection matrices. We compute the standard attention map $\mathbf{A} \in \mathbb{R}^{B \times H \times N \times N}$ via scaled dot-product:

$$\mathbf{A} = \text{Softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right). \quad (3)$$

As depicted in Fig. 3b, SGSA introduces a lightweight pruning mechanism guided by a hybrid metric to rigorously filter non-informative tokens. To measure the importance of the i th token, we formulate a comprehensive score S_i integrating relational saliency from attention weights with informational saliency from feature magnitudes:

$$S_i = \alpha \cdot \underbrace{\text{Norm} \left(\frac{1}{H} \sum_{h=1}^H \max_j \mathbf{A}_{h,i,j} \right)}_{S_i^{\text{rel}}: \text{Relational Saliency}} + (1 - \alpha) \cdot \underbrace{\text{Norm}(\|\mathbf{q}_i\|_2)}_{S_i^{\text{info}}: \text{Informational Saliency}} \quad (4)$$

where H is the number of heads, $\mathbf{q}_i$ is the query vector, and $\|\cdot\|_2$ denotes the L2-norm. Here, S_i^{rel} aggregates the maximum attention intensity token i assigns to any key, reflecting its role in capturing global dependencies. Concurrently, S_i^{info} leverages the feature vector L2-norm, which strongly correlates with semantic information density. $\text{Norm}(\cdot)$ denotes Min-Max normalization and α is a balancing coefficient.

Based on scores S , we identify the indices of the Top- K most salient tokens $\mathbf{I}_{\text{top}} = \text{TopK}(S, K)$. It is worth noting that the Top- K token pruning is performed globally rather than per-head, ensuring that all attention heads share an identical set of unpruned spatial indices to maintain structural consistency. The attention operation is then restricted exclusively to these selected tokens:

$$\mathbf{O}_{\text{sparse}} = \text{Softmax} \left(\frac{\mathbf{Q}_{\mathbf{I}_{\text{top}}} \mathbf{K}^T}{\sqrt{d}} \right) \mathbf{V} \quad (5)$$

A direct residual connection between sparse output $\mathbf{O}_{\text{sparse}} \in \mathbb{R}^{B \times K \times C}$ and dense input $\mathbf{X} \in \mathbb{R}^{B \times N \times C}$ causes a spatial mismatch. To resolve this, the unpooling module in Fig. 3a employs a scatter-and-add mechanism. After processing $\mathbf{O}_{\text{sparse}}$ through the Feed-Forward Network yielding $\mathbf{F}_{\text{sparse}} \in \mathbb{R}^{B \times K \times C}$, we scatter it into a zero-initialized tensor $\mathbf{Z} \in \mathbb{R}^{B \times N \times C}$ using original indices $\mathbf{I}_{\text{top}}$:

$$\mathbf{Z}_{\mathbf{I}_{\text{top}}} = \mathbf{F}_{\text{sparse}}, \quad \mathbf{Z}_{\setminus \mathbf{I}_{\text{top}}} = \mathbf{0}. \quad (6)$$

The residual addition is then safely performed at dense resolution:

$$\mathbf{X}_{\text{out}} = \mathbf{X} + \mathbf{Z}. \quad (7)$$

Thus, salient tokens receive updated contexts while unselected background tokens undergo identity mapping, maintaining strict spatial consistency without redundant computations.

While computing the full attention matrix $\mathbf{A}$ retains $\mathcal{O}(N^2)$ complexity, SGSA's acceleration stems from aggressive token pruning. Despite minor sorting overhead, curtailing the sequence length from N to K significantly reduces the computational load of subsequent Feed-Forward Networks, which dominate standard ViT computation.

3.2.1. Stabilized smoothing strategy

Aggressive token removal during early SGSA training often causes severe information loss and gradient instability. To mitigate this, we devise a stabilized smoothing strategy inspired by reinforcement learning. Progressively adjusting sampling ratios and injecting noise ensures a stable transition from global learning to selective focusing.

The procedure involves three steps. First, we adopt a progressive sampling rate. During the initial E_w epochs, we utilize a high sampling rate r_{max} to minimize information loss, linearly decaying it to a target base rate r_{base} :

$$r(t) = r_{\text{max}} + (r_{\text{base}} - r_{\text{max}}) \cdot \min \left(1, \frac{t}{E_w} \right), \quad (8)$$

where current epoch t determines the sampled length $L_{\text{out}} = \lfloor T_q \cdot r(t) \rfloor$ for a query sequence of length T_q .

Second, we inject small-scale Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$ before computing token weights w to stabilize their distribution:

$$\tilde{w} = w + \epsilon, \quad \sigma = 0.01. \quad (9)$$

Finally, temperature scaling $\tilde{w} = w/\tau$ with $\tau = 0.1$ adjusts the softmax distribution, controlling Top- K sampling discreteness and preventing overly extreme decisions.

3.3. Geometry-preserving positional embeddings

While SGSA efficiently reduces computational redundancy, its discrete sampling risks disrupting the continuous spatial semantics of the image. To mitigate this, we incorporate Geometry-Preserving Positional Embeddings to explicitly anchor selected tokens to their global contexts. Functioning as rigid geometric anchors, these embeddings inject orthogonal spatial priors across horizontal and vertical orientations. This ensures the model maintains a rigorous awareness of spatial layouts under sparse interactions, robustly capturing directional defect patterns like scratches or cracks. Injected homogeneously into query, key, and value tokens, they naturally collaborate with the S^3 -Block to guide attention with precise structural integrity.

To implement these priors, we adopt a sinusoidal encoding mapping spatial coordinates into high-dimensional features. For a feature map of resolution $H \times W$, we generate normalized vertical and horizontal coordinates denoted $\bar{y}_i$ and $\bar{x}_j$. These coordinates are scaled to $[0, 2\pi]$ ensuring sufficient phase shift sensitivity. To capture multi-scale structures, they are transformed using varying frequencies. Formally, for a coordinate $p \in \{\bar{y}_i, \bar{x}_j\}$, the encoding function $\phi(p)$ at channel dimensions $2k$ and $2k + 1$ is:

$$\begin{aligned} \phi(p, 2k) &= \sin(p/\Omega^{2k/D_{\text{pos}}}), \\ \phi(p, 2k + 1) &= \cos(p/\Omega^{2k/D_{\text{pos}}}), \end{aligned} \quad (10)$$

where temperature parameter $\Omega = 10000$ controls the frequency spectrum, and D_{pos} is the dimension per spatial axis. Specifically, $D_{\text{pos}} =$

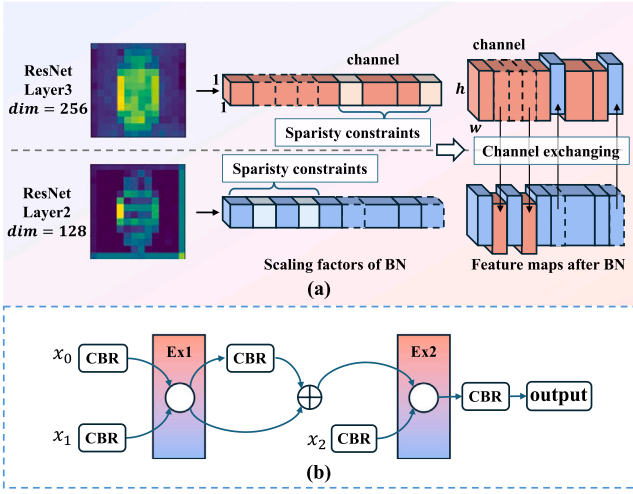


Fig. 4. Overall structure of the Exchangeable FPN. (a): Cross-scale exchange module improving multi-scale feature integration without increasing parameters. (b): Main architecture of the Exchangeable FPN, where CBR denotes Conv+BN+ReLU.

$D/2$, with D representing the total embedding dimension. The separate axis encodings are then concatenated into the definitive embedding $\mathbf{P}_{(i,j)} \in \mathbb{R}^D$:

$$\mathbf{P}_{(i,j)} = \text{Concat} [\phi(\bar{y}_i), \phi(\bar{x}_j)]. \quad (11)$$

This explicitly encodes absolute spatial configurations, enabling SGSA to maintain rigorous geometric alignment during sparse interactions.

3.4. Exchangeable feature pyramid network

Naive element-wise summation in traditional FPNs causes semantic confusion and channel redundancy, obscuring fine-grained anomaly details. We therefore propose the Exchangeable FPN, a parameter-efficient fusion module inspired by Channel Exchange Networks [14] and AlexNet [44]. By utilizing normalized Batch Normalization scaling coefficients as content-aware gates, this module enables adaptive cross-scale channel exchange. This integrates global semantics with local details without additional parameter overhead, significantly enhancing localization capabilities as depicted in Fig. 4.

Let inputs be three feature maps x_0, x_1, x_2 with respective channel dimensions d_0, d_1, d_2 . We first generate an intermediate representation from the lowest-scale feature via a convolution and adaptation layer:

$$\text{fpn}_0 = \text{ReLU}(\text{BN}_1(\text{Conv}_1(x_0))), \quad (12)$$

where Conv_1 is a 3×3 convolution with stride 2. Similarly, we adapt x_1 to the intermediate dimension:

$$x'_1 = \text{ReLU}(\text{BN}'_1(\text{Conv}'_1(x_1))), \quad (13)$$

where Conv'_1 is a 1×1 convolution adjusting the channel count.

Taking fpn_0 and x'_1 as branches, we apply the exchange module using Batch Normalization scaling vectors γ and a threshold θ . This generates masks $m_c = \mathbb{I}(|\gamma_c| \geq \theta)$ to perform channel exchange for low-importance channels:

$$\mathbf{y}_c^{(1)} = m_c^{(1)} \mathbf{x}_c^{(1)} + (1 - m_c^{(1)}) \mathbf{x}_c^{(2)}, \quad \mathbf{y}_c^{(2)} = m_c^{(2)} \mathbf{x}_c^{(2)} + (1 - m_c^{(2)}) \mathbf{x}_c^{(1)}. \quad (14)$$

Post-exchange, we fuse features to obtain fpn_1 and repeat this process for x_2 . Final outputs are processed by ResNet blocks and an FPN. This design mitigates information confusion and preserves discriminative channels, improving anomaly detection accuracy and efficiency.

4. Experiment

4.1. Datasets and evaluation metrics

Datasets. We evaluate $S^3\text{AD}$ on four benchmarks assessing diverse anomaly complexities. For established industrial scenarios, we utilize MVTec AD with 15 categories across 5354 images alongside VisA containing 12 categories across 10,821 images. To rigorously test scalability and generalization, we employ Real-IAD, the largest industrial dataset comprising 30 objects and over 150,000 images. Finally, we include MPDD featuring 6 metal categories and 1346 images to evaluate robustness, as its small, low-contrast defects pose significant challenges to fine-grained localization.

Evaluation Metrics. Following standard protocols [4,9], for anomaly detection and segmentation, we report Area Under the Receiver Operating Characteristic Curve (AU-ROC), Average Precision (AP), and F1-score-max (F1-max). Additionally, for anomaly segmentation, we report Area Under the Per-Region-Overlap (AU-PRO). We further calculate the mean of these seven evaluation metrics (denoted as mAD) to represent a model's comprehensive capability. Results are reported as the mean $\pm$ standard deviation across three independent runs.

4.2. Implementation details

We implement $S^3\text{AD}$ in PyTorch 2.8.0 on an Ubuntu workstation equipped with an NVIDIA RTX 4090 GPU. Input images are resized to 256×256 , and the teacher network utilizes an ImageNet-pre-trained ResNet-34 backbone. Models are optimized using AdamW with a 10^{-3} learning rate, 10^{-4} weight decay, and $\beta = (0.9, 0.999)$ at a batch size of 20. The objective is pixel-wise Mean Squared Error. Training includes a 3-epoch warmup, with evaluations every 20 epochs after epoch 1000. To stabilize SGSA sampling, a drop-rate annealing strategy progressively decays the sampling rate during warmup.

For reproducibility, core hyperparameters are explicitly defined. The SGSA hybrid saliency weight α is set to 0.7. During stabilized smoothing, the Top-K retention ratio linearly decays from an initial maximum $r_{\max} = 0.8$ to a base ratio $r_{\text{base}} = 0.5$ over a 10-epoch warmup E_w . Concurrently, noise injection magnitude σ is 0.01 and temperature scaling τ is 0.1 to prevent premature convergence. Finally, the Exchangeable FPN channel exchange threshold θ is 0.02.

4.3. Comparative experiments on different datasets

Compared methods. We evaluate against primary baselines including reconstruction-based RD [45] and UniAD [8], alongside embedding-based SimpleNet [19]. As these originally report single-class results, we reproduce their MUAD performance via the ADER platform.

In addition, we compare several existing MUAD methods, including the reconstruction-based MambaAD and ViTAD methods, the DecAD method [24] that combines data augmentation and reconstruction, and the DiAD method [23] which employs diffusion models, to thoroughly verify the effectiveness of our approach.

4.3.1. MVTec AD

Quantitative Performance. As detailed in Table 2, $S^3\text{AD}$ achieves state-of-the-art performance on MVTec AD, systematically surpassing baselines like MambaAD and ViTAD. At the image level, $S^3\text{AD}$ attains perfect Image-AUC scores (100.0) on seven categories (e.g., Carpet, Metal Nut), demonstrating strong robustness. At the pixel level, it excels in fine-grained localization, delivering significant improvements in Pixel-AUC and Pixel-PRO on challenging categories like Cable and Transistor.

Aggregated Metric and Analysis. $S^3\text{AD}$ reaches a highly competitive mAD of 86.6, alongside superior average Image-AUC (99.1) and Pixel-AUC (98.0). This pronounced advantage on structurally complex objects suggests that the SGSA mechanism effectively handles complex semantics and background interference, robustly calibrating decision boundaries without sacrificing segmentation precision.

Table 1
Comparative results of different methods on multiple datasets.

Dataset	Methods	Source	Image-level			Pixel-level				mAD
			AUROC	AP	$F_{1,max}$	AUROC	AP	$F_{1,max}$	AUPRO	
MVTec-AD	RD	CVPR22	94.6	96.5	95.2	96.1	48.6	53.8	91.1	82.3
	UniAD	NIPS22	96.5	98.8	96.2	96.8	43.4	49.5	90.7	81.7
	SimpleNet	CVPR23	95.3	98.4	95.8	96.9	45.9	49.7	86.5	81.2
	DiAD	AAAI24	97.2	99.0	96.5	96.8	52.6	55.5	90.7	84.0
	MambaAD	NIPS24	98.6	99.6	97.8	97.7	56.3	59.2	93.1	86.0
	ViTAD	CVIU25	98.9	99.6	98.2	98.1	57.1	59.6	94.4	86.6
	DecAD	ICCV25	99.0	99.7	98.3	97.8	60.7	62.5	92.5	87.2
	Our	-	99.1$\pm$0.1	99.7$\pm$0.0	98.6$\pm$0.1	98.0$\pm$0.1	57.5$\pm$0.2	60.1$\pm$0.2	93.2$\pm$0.2	86.6$\pm$0.1
VisA	RD	CVPR22	92.4	92.4	89.6	98.1	38.0	42.6	91.8	77.8
	UniAD	NIPS22	88.8	90.8	85.8	98.3	33.7	39.0	85.5	74.6
	SimpleNet	CVPR23	87.2	87.0	81.8	96.8	34.7	37.8	81.4	72.4
	DiAD	AAAI24	86.8	88.3	85.1	96.0	26.1	33.0	75.2	70.1
	MambaAD	NIPS24	94.3	94.5	89.4	98.5	39.4	44.0	91.0	78.7
	ViTAD	CVIU25	90.3	91.2	86.4	98.2	36.4	40.9	85.8	75.6
	DecAD	ICCV25	94.7	95.5	90.3	98.5	43.6	47.3	91.7	80.2
	Our	-	94.9$\pm$0.1	95.3$\pm$0.1	91.1$\pm$0.1	99.0$\pm$0.0	39.3$\pm$0.0	44.0$\pm$0.3	92.3$\pm$0.1	79.4$\pm$0.0
Real-IAD	RD	CVPR22	82.7	79.3	74.1	97.2	25.2	32.8	90.0	68.8
	UniAD	NIPS22	83.1	81.2	74.5	97.4	23.3	30.9	87.1	68.2
	SimpleNet	CVPR23	54.9	50.6	61.5	76.1	1.9	4.9	42.4	41.8
	DiAD	AAAI24	75.6	66.4	69.9	88.0	2.9	7.1	58.1	52.6
	MambaAD	NIPS24	<u>87.0</u>	<u>85.3</u>	<u>77.6</u>	98.6	32.4	38.1	<u>91.2</u>	72.9
	ViTAD	CVIU25	82.7	80.2	73.7	97.2	24.3	32.3	84.8	67.9
	DecAD	ICCV25	85.4	83.7	76.1	98.3	36.7	42.5	89.0	<u>73.1</u>
	Our	-	88.3$\pm$0.1	86.5$\pm$0.0	79.2$\pm$0.0	98.5$\pm$0.0	35.4$\pm$0.5	40.9$\pm$0.4	92.0$\pm$0.0	74.4$\pm$0.2
MDPP	RD	CVPR22	90.3	92.8	90.5	98.3	39.6	40.6	95.2	78.2
	UniAD	NIPS22	80.1	83.2	85.1	95.4	19.0	25.6	83.8	67.5
	SimpleNet	CVPR23	90.6	94.1	89.7	97.1	33.6	35.7	90.0	75.8
	DiAD	AAAI24	85.8	89.2	86.5	91.4	15.3	19.2	66.1	64.8
	MambaAD	NIPS24	<u>89.2</u>	<u>93.1</u>	<u>90.3</u>	97.7	33.5	<u>38.6</u>	92.8	<u>76.5</u>
	ViTAD	CVIU25	87.4	90.8	87.0	97.8	34.6	37.8	92.9	75.5
	Our	-	94.2$\pm$0.3	96.6$\pm$0.1	93.0$\pm$0.3	98.1$\pm$0.0	41.7$\pm$0.4	43.5$\pm$0.5	93.9$\pm$0.0	80.1$\pm$0.2

Table 2
Multi-class anomaly detection results with Image-AUC/Pixel-AUC/Pixel-PRO metrics and mAD on MVTec AD.

Method $\rightarrow$		UniAD (You et al., 2022a)	MambaAD (He et al., 2024a)	ViTAD (Zhang et al., 2025)	S^3AD
Category $\downarrow$		NIPS22	NIPS24	CVIU25	Ours (Mean $\pm$ Std)
Texture	Carpet	99.8/ 98.4/ 94.4	99.8/ 99.2/ 97.3	99.5/ 99.0/ 94.7	100.0$\pm$0.0/99.2$\pm$0.0/97.0$\pm$0.2
	Grid	99.3/ 97.7/ 92.9	99.8/ 98.9/ 96.9	99.7/ 98.6/ 95.8	100.0$\pm$0.0/99.3$\pm$0.0/96.9$\pm$0.5
	Leather	100./ 98.8/ 96.8	100./ 99.3/ 98.7	100./ 99.6/ 97.9	100.0$\pm$0.0/99.5$\pm$0.0/98.6$\pm$0.1
	Tile	99.9/ 92.3/ 78.4	96.8/ 93.0/ 79.5	100./ 96.6/ 87.0	99.1$\pm$0.1/94.8$\pm$0.2/82.4$\pm$0.9
	Wood	98.9/ 93.2/ 86.7	98.5/ 94.0/ 92.0	98.7/ 96.4/ 88.0	97.6$\pm$0.1/94.6$\pm$0.4/89.4$\pm$0.5
Object	Bottle	100./ 98.0/ 93.1	99.8/ 98.7/ 96.0	100./ 98.8/ 94.3	100.0$\pm$0.0/98.7$\pm$0.0/95.7$\pm$0.1
	Cable	94.8/ 96.9/ 86.1	99.3/ 95.2/ 90.4	98.5/ 96.2/ 90.2	99.4$\pm$0.1/98.4$\pm$0.1/92.8$\pm$0.2
	Capsule	93.7/ 98.8/ 92.1	91.8/ 98.3/ 93.0	95.4/ 98.3/ 92.0	97.8$\pm$0.4/98.8$\pm$0.0/95.1$\pm$0.1
	Hazelnut	100./ 98.0/ 94.1	100./ 99.1/ 95.3	99.8/ 99.0/ 95.2	100.0$\pm$0.0/99.1$\pm$0.0/95.7$\pm$0.1
	Metal Nut	98.3/ 93.3/ 81.8	99.7/ 96.7/ 92.9	99.7/ 96.4/ 92.4	100.0$\pm$0.0/96.8$\pm$0.1/92.8$\pm$0.1
	Pill	94.4/ 97.5/ 95.8	94.7/ 96.2/ 95.2	96.2/ 98.7/ 95.3	98.5$\pm$0.0/98.3$\pm$0.0/96.6$\pm$0.0
	Screw	95.3/ 99.2/ 95.2	92.7/ 99.3/ 97.0	91.3/ 99.0/ 93.5	97.5$\pm$0.6/99.7$\pm$0.0/98.2$\pm$0.1
	Toothbrush	89.7/ 98.4/ 87.9	96.4/ 98.9/ 92.0	98.9/ 99.1/ 90.9	99.4$\pm$0.3/98.9$\pm$0.0/91.5$\pm$0.1
	Transistor	99.8/ 97.4/ 93.5	99.9/ 96.0/ 90.4	98.8/ 93.9/ 76.8	100.0$\pm$0.0/96.9$\pm$0.3/91.5$\pm$0.6
	Zipper	98.6/ 98.0/ 92.6	98.2/ 98.1/ 94.5	97.6/ 95.9/ 87.2	99.9$\pm$0.0/98.7$\pm$0.0/95.7$\pm$0.0
Average		97.5/ 97.0/ 90.7	97.8/ 97.4/ 93.4	98.3/ 97.7/ 91.4	99.1$\pm$0.1/98.0$\pm$0.1/93.2$\pm$0.2
mAD		81.7	86.0	86.6	86.6$\pm$0.1

4.3.2. VisA

Quantitative Performance. As detailed in Table 3, S^3AD demonstrates robust generalization on the VisA dataset, which features complex structures and diverse objects. At the image level, our method achieves the highest average Image-AUC of 94.9, consistently outperforming MambaAD (93.9). At the pixel level, S^3AD excels in fine-grained localization, attaining the best average Pixel-AUC (99.0) and Pixel-PRO (92.3). This distinct lead in the localization-sensitive Pixel-PRO indicates that S^3AD effectively maintains boundary consistency and filters background noise across challenging scenarios.

Aggregated Metric and Analysis. Regarding the comprehensive mAD metric, S^3AD achieves a leading score of 79.4, solidifying its

superiority over MambaAD (78.7) and ViTAD (75.6). Without relying on heavy augmentation pipelines, S^3AD effectively handles high intra-class variance and complex multi-instance objects. These results confirm its capability to jointly model multi-scale semantics and fine-grained anomalies for rigorous real-world inspection tasks.

4.3.3. Real-IAD

Quantitative Superiority. Real-IAD presents a rigorous challenge due to its immense scale and realistic distribution shifts. Detailed in Table 1, S^3AD demonstrates strong robustness by establishing a new state-of-the-art aggregated mAD of 74.43. This distinctly outperforms DecAD at 73.1, MambaAD at 72.9, and ViTAD at 67.9. At the image

Table 3

Multi-class anomaly detection results with Image-AUC/Pixel-AUC/Pixel-PRO metrics and mAD on VisA.

Method → Category ↓		Image-AUC / Pixel-AUC / Pixel-PRO			
		UniAD (You et al., 2022a)	MambaAD (He et al., 2024a)	ViTAD (Zhang et al., 2025)	S ³ AD
		NIPS22	NIPS24	CVIU25	Ours(Mean±Std)
Complex Structure	PCB1	94.2/ 99.2/ 88.8	96.0/ 99.8 / 95.1	95.8/ 99.5/ 89.6	98.2±0.1/99.8±0.0/96.3±0.2
	PCB2	91.1/ 98.0/ 82.2	96.1/ 99.0/ 91.2	90.6/ 97.9/ 82.0	98.1±0.2/99.2±0.0/92.8±0.3
	PCB3	82.2/ 98.2/ 79.3	95.2/ 99.1/ 92.5	90.9/ 98.2/ 88.0	97.5±0.0/99.2±0.0/93.0±0.1
	PCB4	99.0/ 97.2/ 82.9	99.8/ 98.6/ 90.7	99.1/ 99.1 / 91.8	99.9±0.0/98.9±0.0/91.8±0.2
Multiple Instances	Macaroni1	82.8/ 99.0/ 92.6	92.0/ 99.5/ 96.3	85.8/ 98.5/ 89.2	93.6±0.4/99.7±0.0/97.1±0.1
	Macaroni2	76.0/ 97.3/ 87.0	82.0/ 99.4/ 96.4	79.1/ 98.1/ 87.2	85.3±0.4/99.5±0.0/97.1±0.1
	Capsules	70.3/ 97.4/ 72.2	89.8/ 99.2/ 93.2	79.2/ 98.2/ 75.1	88.4±0.2/99.2±0.0/92.7±0.4
	Candle	95.8/ 99.0/ 93.0	96.9 / 99.1/ 96.1	90.4/ 96.2/ 85.2	96.6±0.2/99.2±0.0/96.1±0.0
Single Instance	Cashew	94.3/ 99.0 / 88.5	95.3 / 94.9/ 89.3	87.8/ 98.5/ 78.8	93.5±0.2/98.7±0.0/90.8±0.1
	Chewing Gum	97.5/ 99.1 / 85.0	97.8/ 98.2/ 80.7	94.9/ 97.8/ 71.5	98.6±0.1/98.5±0.0/82.1±0.2
	Fryum	86.9/ 97.3/ 82.0	96.3/ 97.1/ 92.4	94.3/ 97.5 / 87.8	96.6±0.2/99.3±0.0/92.0±0.2
	Pipe Fryum	95.3/ 99.1/ 93.0	98.8/ 99.1/ 95.7	97.8/ 99.5 / 94.7	99.7±0.0/99.2±0.0/95.7±0.2
Average		88.8/ 97.0/ 90.7	93.9/ 98.5/ 92.1	90.5/ 98.2/ 85.1	94.9±0.1/99.0±0.0/92.3±0.1
mAD		74.5	78.7	75.6	79.4±0.0

Table 4End-to-End Inference Performance Comparison. The top 3 results in each column are highlighted in **dark**, medium, and light green, respectively. “Tiny” and “Small” versions use decoder depths of [2, 2, 2, 2] and [3, 6, 6, 3].

Model	Source	Efficiency Metrics					Performance Metrics		
		Throughput↑ (FPS)	Latency↓ (ms)	FLOPs↓ (G)	Params↓ (M)	Memory↓ (MB)	mAD (Mvtec)↑	mAD (VisA)↑	mAD (MPDD)↑
UniAD	NIPS22	44.84	22.30	3.36	24.52	125.27	81.70	74.60	67.50
MambaAD	NIPS24	29.40	34.01	8.30	25.67	201.22	86.00	78.70	76.50
ViTAD	CVIU25	108.89	9.18	9.67	38.59	163.45	86.60	75.60	75.50
S ³ AD-tiny	Ours	31.52	31.72	6.79	25.60	143.15	85.90	78.10	80.30
S ³ AD-small	Ours	16.19	61.75	9.41	37.86	190.97	86.40	79.10	80.70
S ³ AD	Ours	11.63	85.99	12.03	43.36	212.66	86.70	79.40	80.14

level, S³AD surpasses all competitors across AUROC, AP, and F1-max. At the pixel level, our method delivers an 11% mAP gain over ViTAD and consistently exceeds DecAD in AU-PRO structural consistency, proving superior precise boundary delineation.

Mechanism Analysis. These results validate SGSA’s efficacy in complex environments. Unlike standard approaches struggling with high-redundancy backgrounds, SGSA effectively filters irrelevant noise, focusing computational resources solely on salient defect patterns. This enables S³AD to maintain high localization precision under heavy interference, confirming its robustness and scalability for real-world deployment.

4.3.4. MPDD

Quantitative Superiority. The MPDD dataset challenges models with painted metal surfaces and multi-scale fine structures. S³AD establishes a substantial lead with an aggregated mAD of 80.15, clearly outperforming MambaAD at 76.5 and ViTAD at 75.5. At the image level, our method improves AUROC by 6.62% and 4.82% against ViTAD and MambaAD respectively, proving robustness in distinguishing defects across varying backgrounds. At the pixel level, S³AD surpasses MambaAD by over 8.4% in mAP and 5.0% in F1-max, confirming superior precision in segmenting fine-grained defects frequently missed by global scanning methods.

Mechanism Analysis. Success on MPDD validates S³AD’s architectural synergy. Strong localization stems from the Exchangeable FPN and Geometry-Preserving PE collaboratively ensuring intricate structural details are preserved and fused across scales. By effectively handling these spatial semantics, S³AD demonstrates strong cross-category generalization for high-precision inspection of complex components.

4.3.5. End-to-end efficiency analysis

Beyond accuracy, industrial deployment imposes strict real-time throughput and memory constraints. Table 4 systematically compares S³AD against recent state-of-the-art models evaluating this trade-off.

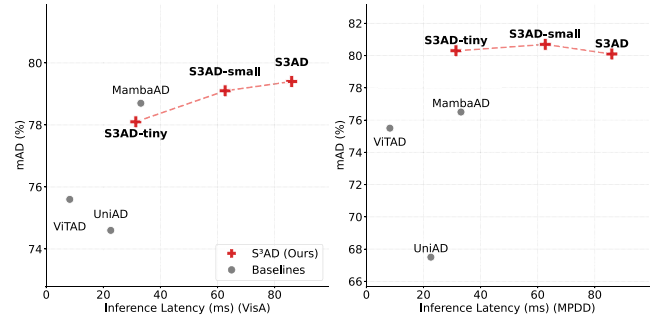


Fig. 5. Efficiency–accuracy Pareto plots on VisA and MPDD datasets. The dashed line connects our S³AD variants (tiny, small, base), demonstrating a superior scalability and trade-off landscape compared to existing baselines.

Our architecture establishes a highly competitive Pareto frontier visualized in Fig. 5, occupying the top-right region across varying computational budgets. Specifically, the lightweight S³AD-tiny variant significantly curtails theoretical FLOPs to 6.79 G and peak GPU memory to 143.15 MB compared to MambaAD and ViTAD, while maintaining a robust throughput exceeding 30 FPS. Furthermore, our compact S³AD-small and S³AD-tiny variants exhibit compelling performance, achieving competitive detection precision on VisA and MPDD benchmarks. These results underscore a superior accuracy–efficiency trade-off, proving our sparsity-driven paradigm effectively satisfies the stringent constraints of real-world deployment.

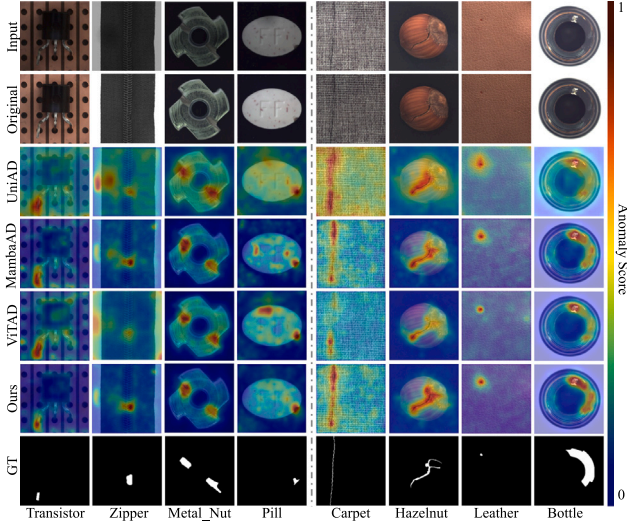
4.3.6. Robustness under illumination variations

Quantitative Results. As shown in Table 5, S³AD maintains strong robustness under both *shadow* and *spot* illumination conditions. On the *Shadow-augmented* MvtecAD, our method achieves the best overall

Table 5

Comparative results of different methods on the Shadow- and Spot-augmented MVTec AD dataset.

Dataset	Methods	Image-level			Pixel-level				mAD
		AUROC	AP	$F_{1,max}$	AUROC	AP	$F_{1,max}$	AUPRO	
Shadow	UniAD	89.2	95.6	94.8	95.1	35.8	42.5	86.1	77.0
	MambaAD	95.9	98.5	96.7	95.6	52.4	56.3	88.2	83.4
	VITAD	91.8	96.6	94.8	96.5	48.3	51.7	88.0	81.1
	S3AD	96.0	98.7	96.8	97.4	51.7	54.8	90.7	83.7
Spot	UniAD	92.9	97.6	95.1	95.1	39.0	45.6	86.6	78.8
	MambaAD	95.9	98.5	96.7	95.6	52.4	56.3	88.2	83.4
	VITAD	94.1	97.8	95.6	94.7	49.4	52.1	83.6	81.0
	S3AD	95.3	98.6	97.2	96.5	51.5	55.7	87.8	83.2

**Fig. 6.** Qualitative comparison of anomaly localization results under shadow and spot illumination conditions.

performance, obtaining the highest mAD and leading results on several image-level and pixel-level metrics. On the *Spot-augmented* MVTecAD, S³AD also remains highly competitive, demonstrating stable detection capability under strong local illumination interference. These results indicate that the proposed SGSA with hybrid-metric Top-K pruning is robust to illumination variations.

Qualitative Analysis. Fig. 6 further illustrates the robustness of S³AD under different lighting conditions. Under *shadow* interference, our method effectively suppresses false responses caused by shaded regions, while the compared methods show noticeable misclassification on samples such as *Zipper* and *Pill*. Under the *spot* setting, S³AD preserves stronger sensitivity to subtle defects and produces clearer localization on samples such as *Carpet* and *Hazelnut*, where other methods tend to miss crack regions under strong light. These visual results further confirm the superior robustness of our method to illumination interference.

4.3.7. Visualization results

Fig. 7 visualizes anomaly maps across diverse industrial scenarios, highlighting four key advantages of S³AD:

Precision on Single Objects. For distinct objects such as bottles and cables, S³AD exhibits superior focus on defective regions. Unlike baselines producing diffuse noise or ambiguous boundaries, our anomaly maps align tightly with ground-truth masks. This validates SGSA effectively suppresses irrelevant background context, pinpointing exact anomaly locations rather than merely flagging their presence.

Robustness on Texture-Rich Surfaces. On large-area textured samples like grids, S³AD demonstrates exceptional stability. While RD and UniAD suffer high false-positive rates from background interference and MambaAD yields scattered predictions, S³AD accurately

traces defect contours without hypersensitivity to repetitive patterns. This confirms our sparsity-driven strategy strongly filters redundant texture information.

Boundary Clarity in Multi-Instance Scenes. In scenarios with adjacent objects including candles and capsules, S³AD maintains rigorous decision boundaries. While methods like RD and ViTAD exhibit signal leakage into neighboring normal instances, our approach successfully isolates specific anomalies. This spatial coherence underscores Geometry-Preserving PE's efficacy in maintaining structural awareness across complex layouts.

Fine-Grained Localization. On dense structures like PCBs, S³AD excels in pinpointing subtle abnormalities. Contrasting with the blurry responses of competing methods, our model highlights fine-grained defects around chips and solder joints with minimal noise. This high-fidelity localization demonstrates sensitivity to minute structural deviations, offering superior interpretability for real-world deployment.

4.4. Ablation study

We systematically ablate the core components of S³AD — SGSA, Geometry-Preserving PE, E-FPN, and the Smoothing strategy — using the VisA PCB subset detailed in Table 6.

Individual contributions. The baseline model achieves near-saturated image-level performance with AUROC and AP exceeding 98%. Despite this, introducing any single component yields consistent mAD gains from a baseline of 81.25. Specifically, integrating SGSA, PE, and E-FPN individually raises the mAD to 81.29, 81.60, and 81.66 respectively. This confirms multi-scale feature interaction and explicit spatial modeling positively contribute to the detection pipeline.

Backbone and Decoder Depth. Detailed in Table 7, increasing decoder depth consistently improves mAD across pre-trained backbones. Notably, ResNet34 with a [3, 9, 9, 7] configuration achieves an optimal efficiency–accuracy trade-off. It matches the peak performance of the significantly heavier WideResNet50 while requiring merely a fraction of its parameters and FLOPs.

Synergistic analysis. Integrating multiple components exhibits a clear synergistic effect rather than simple additive gains. Combining SGSA, PE, and E-FPN boosts the mAD to 81.71. Furthermore, incorporating the Smoothing strategy elevates the mAD to a peak of 81.86, achieving the best overall performance.

Metric interpretation. Due to near-saturated image-level baseline performance, module benefits primarily manifest in pixel-level metrics like AUPRO and the overall mAD. As PCB subsets contain fine-grained anomalies and complex textures, these consistent gains suggest the proposed modules significantly enhance localization robustness and precision in challenging scenarios, rather than merely making conservative sample-level decisions.

4.4.1. Impact of SGSA

Efficacy and Efficiency. Fig. 8 shows SGSA precisely selects salient regions and suppresses benign textures. Table 6 details this raises baseline mAD from 81.25 to 81.29, anchoring the architecture's 81.86 peak. Table 11 further confirms SGSA's superior semantic preservation over

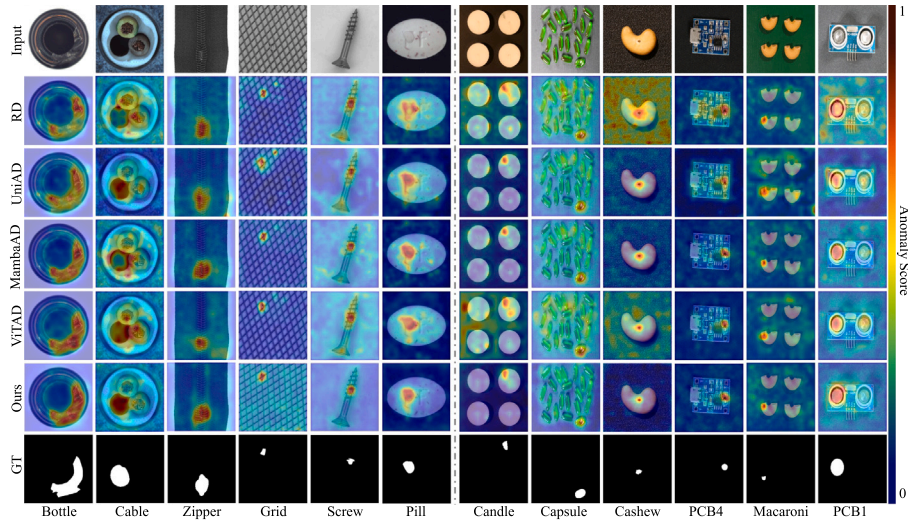


Fig. 7. Qualitative comparison of anomaly localization. Heatmaps are generated via pixel-wise cosine distance between extracted and reconstructed features. S^3 AD demonstrates superior boundary adherence and noise suppression compared to state-of-the-art baselines.

Table 6

Comprehensive ablation results on VisA (PCB subsets averaged). Abbreviations: (A): SGSA, (P): GPPE, (E): Exchange FPN, (S): Smooth.

Method				Image-level			Pixel-level				mAD
A	P	E	S	AUROC	AP	$F_{1,max}$	AUROC	AP	$F_{1,max}$	AUPRO	
✓				98.43	98.29	95.76	99.27	43.36	43.25	92.90	81.25
✓				98.32	98.17	96.10	99.28	41.21	43.14	92.83	81.29 (+0.04)
✓	✓			98.61	98.58	95.83	99.29	42.01	43.78	93.13	81.60 (+0.35)
✓		✓		98.40	98.26	95.96	99.24	40.84	42.46	96.47	81.66 (+0.41)
✓	✓		✓	98.72	98.68	95.84	99.23	42.16	44.39	93.00	81.71 (+0.46)
✓	✓	✓	✓	98.75	98.73	96.34	99.28	42.37	44.32	93.23	81.86 (+0.61)

Table 7

Ablation studies on the pre-trained backbone and S^3 AD decoder depth.

Backbone	Decoder depth	Image-level			Pixel-level				mAD	Params (M)	FLOPs (G)
		AUROC	AP	$F_{1,max}$	AUROC	AP	$F_{1,max}$	AUPRO			
ResNet18	[2, 2, 2, 2]	98.1	99.4	97.2	97.7	54.3	57.5	92.3	85.2	20.2	4.7
	[3, 6, 6, 3]	98.8	99.7	98.0	97.9	55.3	58.9	92.9	85.9	32.5	7.3
	[3, 9, 9, 7]	98.8	99.6	98.1	97.9	55.4	58.9	93.0	86.0	38.0	9.9
ResNet34	[2, 2, 2, 2]	98.5	99.5	98.0	97.8	55.8	58.8	92.8	85.9	25.4	6.6
	[3, 6, 6, 3]	99.0	99.7	98.5	98.0	56.9	59.8	93.0	86.4	37.4	9.0
	[3, 9, 9, 7]	99.1	99.7	98.6	98.0	57.5	60.1	93.2	86.6	42.7	11.3
ResNet50	[3, 9, 9, 7]	98.7	99.5	97.9	97.8	54.6	57.6	92.9	85.6	152.0	37.2
WideResNet50	[3, 9, 9, 7]	99.2	99.8	98.5	98.0	57.2	59.8	93.6	86.6	168.0	44.9

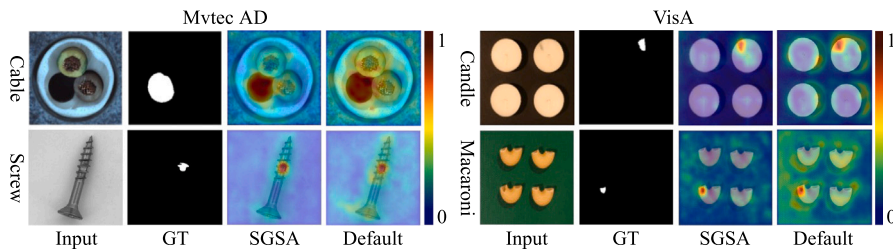


Fig. 8. Qualitative comparison of token selection heatmaps between our SGSA and the Default attention mechanism in MvtecAD and MPDD dataset.

alternative pruning; outperforming ToMe, DyViT, and EViT, it achieves a leading MVTec AD mAD of 86.6. Beyond accuracy, aggressive pruning breaks the efficiency bottleneck. Despite minor sorting overhead at $N = 196$, halving sequences at $N = 4096$ drastically reduces Feed-Forward Network load. Tables 8 and 9 profile a 1.83× block-level speedup and

over 10% end-to-end FPS gain. Scaling to 1024×1024 , this sparse design mitigates standard attention's quadratic memory explosion for sustainable high-resolution deployment.

Hybrid Weight α Sensitivity. The coefficient α in Eq. (4) balances relational saliency against informational density. Ablating $\alpha \in$

Table 8

Attention-level Inference Latency(ms) across varying sequence lengths and batch sizes. We compare the dense Baseline Attention with our proposed token-pruned SGSA. Green text highlights the settings where our sparse mechanism achieves an attention-level speedup.

Batch Size	Seq ($N = 196$)			Seq ($N = 1024$)			Seq ($N = 4096$)		
	Base	Ours	Speedup	Base	Ours	Speedup	Base	Ours	Speedup
1	0.335	0.600	0.56×	0.459	0.579	0.79×	3.697	2.023	1.83×
4	0.376	0.545	0.69×	1.734	1.771	0.98×	9.824	10.208	0.96×
8	1.392	0.578	2.41×	3.461	3.577	0.97×	21.281	20.216	1.05×
16	0.820	0.657	1.25×	6.935	6.764	1.03×	29.052	30.263	0.96×
24	1.186	0.963	1.23×	10.315	10.116	1.02×	59.570	62.952	0.95×

Table 9

System-level efficiency comparison between the Baseline and our proposed SGSA across different input resolutions. Bold values indicate superior performance.

Resolution	Model	FPS ($\uparrow$)	GFLOPs ($\downarrow$)	Params (M) ($\downarrow$)	Mem (MB) ($\downarrow$)	Lat (ms) ($\downarrow$)
256×256	S ³ AD (Attn)	9.22	11.40	30.93	170.10	108.45
	S ³ AD (SGSA)	10.10	12.03	43.36	216.03	99.01
512×512	S ³ AD (Attn)	8.65	44.12	30.93	295.73	115.60
	S ³ AD (SGSA)	10.30	46.18	43.36	330.91	97.13
1024×1024	S ³ AD (Attn)	5.96	174.97	30.93	798.23	167.77
	S ³ AD (SGSA)	6.96	182.78	43.36	783.16	143.72

Table 10

Ablation study results of the SGSA hybrid weight α on the MVTec AD dataset.

α	Image-level			Pixel-level				mAD
	AUROC	AP	$F_{1,max}$	AUROC	AP	$F_{1,max}$	AUPRO	
0	98.6	99.6	98.3	97.9	57.9	60.4	93.2	86.5
0.3	99.1	99.7	98.7	98.0	57.8	60.4	93.2	86.7
0.5	99.1	99.7	98.6	98.0	57.5	60.3	93.1	86.6
0.7	99.2	99.8	98.6	98.0	57.8	60.3	93.2	86.7
1.0	99.1	99.7	98.5	97.9	57.7	60.1	93.1	86.6

{0, 0.3, 0.5, 0.7, 1.0} on MVTec AD (Table 10) reveals relying exclusively on feature magnitudes or attention scores yields suboptimal results. Despite demonstrating remarkable robustness with mAD fluctuating merely 0.2% across all settings, $\alpha = 0.7$ provides the optimal balance. Achieving the highest image-level AUROC of 99.2 alongside a peak mAD of 86.7, it is adopted as the default configuration.

4.4.2. Impact of geometry-preserving PE

Although SGSA optimizes efficiency via dynamic token selection, its discrete sampling inevitably fragments continuous spatial semantics. To restore geometric fidelity, we incorporate Geometry-Preserving Positional Encoding (GP-PE). As evidenced in Table 6, GP-PE provides a critical performance boost, raising mAD to 81.60 (+0.35). This gain significantly exceeds that of adding SGSA alone (+0.04), underscoring the necessity of spatial guidance for fine-grained localization across complex PCB structures. When integrated with E-FPN and Smoothing, the model achieves a peak mAD of 81.86, confirming the robust synergy between direction-sensitive positional information and the overall architecture.

To validate the mechanism, we further evaluate *Equivariance Consistency under Spatial Translation* in Fig. 9. The results confirm that GP-PE effectively anchors the attention mechanism to absolute spatial coordinates, ensuring geometric fidelity during anomaly detection. Furthermore, as compared in Table 12, our GPPE demonstrates robust performance across diverse industrial scenarios. Specifically, GPPE achieves superior performance against both standard Learnable PE (LPE) and Rotary PE (RoPE) on the MVTec AD dataset. On the MPDD dataset, GPPE remains highly competitive, outperforming RoPE and performing closely to LPE.

4.4.3. Impact of exchangeable FPN

Table 13 shows integrating E-FPN yields consistent benchmark gains. With saturated image-level metrics, rising pixel-level AUROC

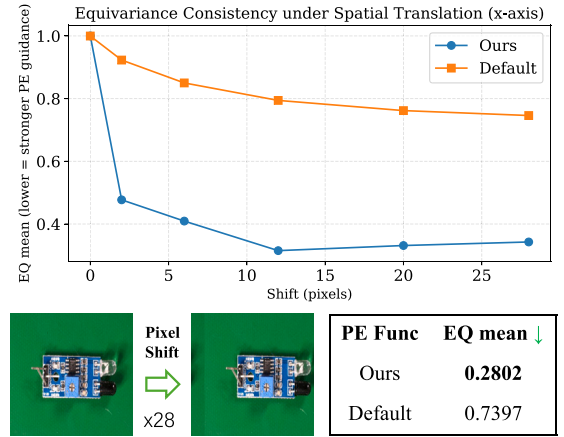


Fig. 9. Visualization of Equivariance Consistency. GP-PE ensures robust spatial anchoring under translation, preserving geometric relationships.

indicates E-FPN primarily enhances fine-grained localization. On VisA-PCB, it emerges as the strongest individual contributor, boosting mAD by 0.41 to 81.66, and peaking at 81.86 alongside Geometry-Preserving PE and Smoothing. Unlike conventional fusion, E-FPN executes parameter-efficient content-aware channel selection before interaction. Table 13 confirms this integration maintains exactly 43.36 M parameters and 12.03 GFLOPs. By suppressing noise during cross-scale fusion, E-FPN improves feature coherence without sacrificing efficiency.

Threshold θ Robustness. We adopt a default threshold of $\theta = 0.02$ inherited directly from Channel Exchange Networks [14] to prevent dataset-specific overfitting. Ablating $\theta \in \{0.02, 0.05, 0.20\}$ on MVTec AD detailed in Table 14 reveals the overall mAD remains remarkably stable, fluctuating by merely 0.1%. This confirms E-FPN is inherently robust to θ variations, proving performance improvements are fundamentally driven by the content-aware channel exchange mechanism rather than strict hyperparameter calibration.

4.4.4. Impact of smoothing strategies

Evaluated on the VisA-PCB subset, the Smoothing Strategy mitigates optimization instability and gradient oscillations induced by abrupt information sparsification (Fig. 10). By synergizing sampling rate annealing, stochastic noise injection, and temperature scaling,

Table 11

Quantitative comparison of different attention and token pruning mechanisms on the MVTec AD dataset.

Attention method	Image-level			Pixel-level				mAD
	AUROC	AP	$F_{1,max}$	AUROC	AP	$F_{1,max}$	AUPRO	
ToMe	66.8	83.0	85.5	79.1	15.1	20.8	60.1	58.6
DyViT	98.9	99.6	98.2	97.9	57.7	60.2	93.2	86.5
EViT	99.0	99.7	98.4	98.0	57.3	60.0	93.1	86.5
SGSA (Ours)	99.1	99.7	98.6	98.0	57.5	60.1	93.2	86.6

Table 12

Ablation study of different Positional Embeddings (PE) on MVTec AD and MPDD datasets. LPE refers to Learnable Positional Embedding, and GPPE represents our proposed Geometry-Preserving Positional Embedding.

PE type	Image-AUC/Pixel-AUC/mAD	
	MVTec AD	MPDD
LPE (Learnable)	99.0/ 98.0 /86.5	94.2 / 98.2 / 80.5
RoPE	98.7/ 98.0 /86.6	92.3/97.6/77.9
GPPE (Ours)	99.1 / 98.0 / 86.7	94.2 / 98.1 / 80.1

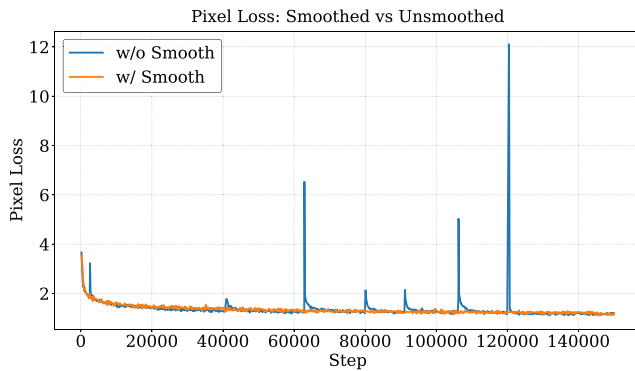


Fig. 10. Comparison of pixel-level loss trajectories. The smoothing strategy mitigates spikes and variance from selective sampling, ensuring stable convergence.

these mechanisms dampen gradient variance and prevent softmax over-sharpening. Quantitatively, this smoother optimization landscape elevates the strong baseline comprising SGSA, PE, and E-FPN from an mAD of 81.71 to a peak of 81.86. Ultimately, stabilizing early-stage training and gradually intensifying selection strength transforms aggressive pruning into robust selectivity, fostering synergy with geometry-preserving components to yield faster convergence and superior localization with negligible overhead.

5. Conclusion and discussion

In this work, we introduce S^3 AD to reconcile the trade-off between global modeling precision and inference efficiency in industrial anomaly detection. Anchored in Selection, Space, and Scale, our framework supplants dense interactions with Saliency-Guided Sparse Attention to filter background redundancy. Synergizing this sparsity with Geometry-Preserving Positional Embeddings and an Exchangeable FPN ensures structural integrity and robust multi-scale fusion. Extensive benchmarking across four authoritative datasets confirms that leveraging visual sparsity yields state-of-the-art accuracy with significantly reduced computational overhead, consistently outperforming strong ViT and Mamba baselines.

Limitations. While S^3 AD effectively balances performance and efficiency, future research will address specific limitations to further optimize the framework. First, replacing the current static Top-K pruning

ratio with a dynamic, adaptive mechanism would better accommodate extreme anomaly scales and highly heterogeneous backgrounds. Second, mitigating the hardware execution overhead introduced by non-contiguous memory access during dynamic token sorting requires specialized low-level optimizations. Finally, transitioning from a uniform Mean Squared Error constraint to spatially-weighted or focal-based distillation objectives could improve the model's sensitivity to subtle defects amidst complex background variations.

Ultimately, S^3 AD demonstrates that intelligently coupling explicit global attention with sparse computation provides a highly effective paradigm for multi-class anomaly detection. By resolving the aforementioned constraints and exploring hardware-aware optimizations, we aim to extend this robust architecture to resource-constrained edge devices, further bridging the gap between academic research and real-time industrial deployment.

CRedit authorship contribution statement

Zhonghua Peng: Writing – original draft, Software, Methodology. **Chengyang Dong:** Writing – original draft, Methodology. **Bo Liu:** Writing – review & editing, Supervision. **Zuoyong Li:** Supervision, Funding acquisition. **Kun Zeng:** Writing – review & editing, Validation. **Jialong Li:** Writing – review & editing, Supervision.

Declaration of Generative AI

The authors used AI tools for language improvement and take full responsibility for the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Table 13

Ablation study of the Exchangeable FPN. The module consistently enhances pixel-level localization across benchmarks while maintaining high image-level detection rates.

Method		Image AUROC/Pixel AUROC			Params (M)	GFLOPs (G)
+MHSA+PE	+Exchange FPN	Mvtec	Visa	MPDD		
✓		99.11/98.07	94.83/98.99	93.84/98.14	43.36	12.03
✓	✓	99.12/98.10	94.83/99.00	94.02/98.16	43.36	12.03

Table 14

Ablation study results of the Exchangeable FPN hyperparameter θ on MVTec AD.

Method (θ)	Image-level			Pixel-level				mAD
	AUROC	AP	$F_{1,max}$	AUROC	AP	$F_{1,max}$	AUPRO	
0.02	99.2	99.8	98.6	98.0	57.8	60.3	93.2	86.7
0.05	99.1	99.7	98.5	98.0	57.4	60.1	93.2	86.6
0.20	99.1	99.7	98.6	98.0	57.6	60.2	93.2	86.6

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